

Number of divisors / sum of divisors

In this article we discuss how to compute the number of divisors $d(n)$ and the sum of divisors $\sigma(n)$ of a given number n .

Number of divisors

It should be obvious that the prime factorization of a divisor d has to be a subset of the prime factorization of n , e.g. $6 = 2 \cdot 3$ is a divisor of $60 = 2^2 \cdot 3 \cdot 5$. So we only need to find all different subsets of the prime factorization of n .

Usually the number of subsets is 2^x for a set with x elements. However this is no longer true, if there are repeated elements in the set. In our case some prime factors may appear multiple times in the prime factorization of n .

If a prime factor p appears e times in the prime factorization of n , then we can use the factor p up to e times in the subset. Which means we have $e+1$ choices.

Therefore if the prime factorization of n is $p_1^{e_1} \cdot p_2^{e_2} \cdot \dots \cdot p_k^{e_k}$, where p_i are distinct prime numbers, then the number of divisors is:

$$d(n) = (e_1 + 1) \cdot (e_2 + 1) \cdot \dots \cdot (e_k + 1)$$

A way of thinking about it is the following:

- If there is only one distinct prime divisor $n = p_1^{e_1}$, then there are obviously $e_1 + 1$ divisors ($1, p_1, p_1^2, \dots, p_1^{e_1}$).
- If there are two distinct prime divisors $n = p_1^{e_1} \cdot p_2^{e_2}$, then you can arrange all divisors in form of a tabular.

$$\begin{array}{cccccc} 1 & p_2 & p_2^2 & \dots & p_2^{e_2} \\ \hline 1 & 1 & p_2 & p_2^2 & \dots & p_2^{e_2} \\ p_1 & p_1 & p_1 & p_1 \cdot p_2 & p_1 \cdot p_2^2 & \dots & p_1 \cdot p_2^{e_2} \\ p_1^2 & p_1^2 & p_1^2 & p_1^2 \cdot p_2 & p_1^2 \cdot p_2^2 & \dots & p_1^2 \cdot p_2^{e_2} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ p_1^{e_1} & p_1^{e_1} & p_1^{e_1} & p_1^{e_1} \cdot p_2 & p_1^{e_1} \cdot p_2^2 & \dots & p_1^{e_1} \cdot p_2^{e_2} \end{array}$$